claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-accdfff4b30936e7e.jsonl`
- SHA-256: `dbd44bca59b11d64a340b6e983894ed31e50ecef8d88be19c7cf4b072201c581`
- Source modified: `2026-05-31T04:25:15+00:00`
- Imported at: `2026-07-05T16:48:20+00:00`
- project: `wf_076beba8-59c`
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`

Transcript

1. user (2026-05-31T04:24:05.606Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):
1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:
- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.
- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Surya's model WEIGHTS are NOT permissively licensed: they are governed by the 'AI PUBS OPEN RAIL-M LICENSE (MODIFIED)', Version 0.1 (March 2, 2023) ? a restricted (OpenRAIL-M family) license, not MIT/BSD/Apache/ISC/HPND. This makes the weights non-compliant with muneem's hard constraint #3 (permissive weights required)."

Source: https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE (primary)
Supporting quote: "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)

Version 0.1, March 2, 2023 (Modified)
http://licenses.ai/"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.
refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.
Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:24:07.987Z)

I'll verify this claim about Surya's model weights license. Let me start by fetching the primary source.

3. assistant (2026-05-31T04:24:12.304Z)

I'll fetch the primary source first to verify the claim about Surya's model weights license.

4. user (2026-05-31T04:24:21.799Z)

Surya Model License Analysis

License Name, Version, and Date:
"AI PUBS OPEN RAIL-M LICENSE (MODIFIED) Version 0.1, March 2, 2023 (Modified)"

First 10 Lines (verbatim):
Lines 1-10 read: "AI PUBS OPEN RAIL-M LICENSE (MODIFIED) / Version 0.1, March 2, 2023 (Modified) / http://licenses.ai/ / PLEASE READ THESE TERMS CAREFULLY BEFORE USING THE MODEL OR A DERIVATIVE WORKS OF THE MODEL MADE AVAILABLE IN CONNECTION WITH THESE TERMS."

Restrictions Present:

Yes, this license contains **both commercial and use-based restrictions**:

- **Commercial restrictions** (Section 2.a-c): Organizations with >$5M gross revenue, >$5M in funding, or competing products require special licensing
- **Behavioral/use-based restrictions** (Attachment A): Prohibits uses that violate laws, infringe IP rights, or create competing products

The license explicitly states these "use restrictions prohibiting the use of the Model" are "intended to avoid potential misuse" and must apply to derivative works.

5. user (2026-05-31T04:24:24.552Z)

License Analysis

First 15 Lines (Verbatim)

"AI PUBS OPEN RAIL-M LICENSE (MODIFIED)

Version 0.1, March 2, 2023 (Modified)
http://licenses.ai/

PLEASE READ THESE TERMS CAREFULLY BEFORE USING THE MODEL OR A DERIVATIVE WORKS OF THE MODEL MADE AVAILABLE IN CONNECTION WITH THESE TERMS. BY DOWNLOADING, REPRODUCING, DISTRIBUTING OR USING THE MODEL OR A DERIVATIVE WORK OF THE MODEL IN ANY MANNER, YOU ("YOU") AGREE TO BE BOUND BY THESE"

License Identification

- **Name:** AI PUBS Open RAIL-M License (Modified)
- **Version:** 0.1
- **Date:** March 2, 2023

Restrictions Present

Yes, this license contains both:

1. **Use-Based/Behavioral Restrictions** (Section 5, Attachment A): The agreement explicitly prohibits usage that violates laws, infringes intellectual property, or breaches the specific conditions outlined in Attachment A.

2. **Commercial Clauses** (Attachment A, Section 2): The license restricts commercial use for entities exceeding certain thresholds?specifically those generating over $5,000,000 in prior-year revenue or having raised equivalent funding. Additionally, it prohibits use by organizations offering competing products or services to the licensor.

These restrictions are enforceable conditions that must be passed along to downstream users (Section 4.a).

6. assistant (2026-05-31T04:24:29.200Z)

The primary source confirms the claim's core assertion. The MODEL_LICENSE file at the cited URL is indeed the "AI PUBS OPEN RAIL-M LICENSE (MODIFIED)", Version 0.1, March 2, 2023 ? exactly as quoted. This is an OpenRAIL-M family license with both commercial revenue thresholds and use-based restrictions.

Now let me check whether this is current/up-to-date and whether the claim might be outdated, since Surya is a fast-moving project. Let me verify the current state of the repo's licensing.

7. user (2026-05-31T04:24:34.649Z)

License Summary for Surya Repository

Code License
Apache 2.0 - The repository uses the standard Apache License 2.0 for the source code.

Model Weights License
The model weights use a **modified AI Pubs Open Rail-M license**. The README states:

> "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue)."

Commercial Licensing
For broader commercial use beyond the startup threshold, the README directs users to:

> "For broader commercial licensing of the model weights, visit our pricing page [here](https://www.datalab.to/pricing)."

The license explicitly mentions the $5M revenue/funding threshold and references OpenRAIL-M as the base license framework for the model weights.

8. user (2026-05-31T04:24:36.091Z)

Web search results for query: "Surya OCR datalab license 2025 commercial OpenRAIL model weights GPL"

Links: [{"title":"surya/MODEL_LICENSE at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/MODEL_LICENSE"},{"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub","url":"https://github.com/datalab-to/surya"},{"title":"surya/README.md at master · datalab-to/surya","url":"https://github.com/datalab-to/surya/blob/master/README.md"},{"title":"GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages","url":"https://github.com/datalab-to/surya/tree/master"},{"title":"Datalab - Intelligent Document Processing: IDP-Software","url":"https://idp-software.com/vendors/datalab/"},{"title":"Surya: The Revolutionary OCR Toolkit for 90+ Languages","url":"https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"},{"title":"surya-ocr 0.17.1 on PyPI - Libraries.io - security & maintenance data for open source software","url":"https://libraries.io/pypi/surya-ocr"},{"title":"pitapo/surya · Hugging Face","url":"https://huggingface.co/pitapo/surya"},{"title":"Datalab develops small AI models to power 'document intelligence' in the enterprise","url":"https://www.forkable.io/p/datalab-develops-small-ai-models"},{"title":"datalab-to/surya-ocr-2 · Hugging Face","url":"https://huggingface.co/datalab-to/surya-ocr-2"}]

Based on the search results, here's what I found regarding Surya OCR's licensing:

Licensing Structure

Surya's model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $2M funding/revenue) and the code is GPL.

Commercial Licensing

For broader commercial licensing or to remove GPL requirements, customers can visit Datalab's pricing page. The GPL code license is paired with a restricted model-weights license that's free below $2M in revenue and paid above that threshold, mirroring a pattern used by other ML infrastructure companies to drive adoption while capturing commercial value.

OpenRAIL License Details

The OpenRAIL-M License is generally applicable to machine-learning models, with "Open" nomenclature indicating that the licensed Model is freely accessible to downstream and other users, while "RAIL" nomenclature indicates that there are use restrictions prohibiting certain uses.

In summary, Surya uses a dual licensing approach: the model weights are under a modified OpenRAIL-M license with free usage for research and startups under $2M revenue, while the code is GPL-licensed with commercial licensing options available for enterprises.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-05-31T04:24:41.131Z)

I have strong corroboration. Let me note one minor discrepancy: there's some variance in sources about whether the code license is GPL or Apache-2.0, and whether the threshold is $2M or $5M. But these details are tangential to the claim under review. The claim is specifically about the MODEL WEIGHTS license being OpenRAIL-M (not permissive), and that is unambiguously confirmed by both the primary source and the README.

Let me verify the exact threshold and check the most current state to ensure the claim isn't outdated, by checking the README directly and the HuggingFace weights page.

10. user (2026-05-31T04:24:45.053Z)

Model Weights License

According to the documentation, the model weights use a **modified AI Pubs Open Rail-M license** with the following terms:

> "free for research, personal use, and startups under $5M funding/revenue"

For broader commercial licensing beyond these permitted uses, the documentation directs users to visit the pricing page at datalab.to/pricing.

The code itself is separately licensed under Apache 2.0, which is more permissive than the model weights license.

11. user (2026-05-31T04:24:47.056Z)

Based on the web page content, the model weights use a **modified AI Pubs Open Rail-M license**.

Here is the relevant quote from the page:

> "The Surya code is licensed under Apache 2.0. The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under $5M funding/revenue). For broader commercial licensing of the model weights, visit our pricing page [here](https://www.datalab.to/pricing?utm_source=gh-surya)."

Additionally, the page lists the license metadata as **"openrail"** in the model information section.

Summary:
- **Code**: Apache 2.0
- **Model Weights**: Modified AI Pubs Open Rail-M (openrail) - free for research, personal use,

and startups under $5M funding/revenue; commercial licensing available separately

12. assistant (2026-05-31T04:24:59.314Z)

The evidence is now conclusive across multiple independent sources, including the HuggingFace model card metadata field which explicitly lists the license as "openrail".

Let me assess the claim against the checklist:

1. **Is the claim supported by the quote?** Yes, exactly. The MODEL_LICENSE file header verbatim matches: "AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1, March 2, 2023." The claim accurately reads this as a non-permissive OpenRAIL-M family license.

2. **Contradicting evidence?** None found. Every source (primary MODEL_LICENSE file, GitHub README, HuggingFace model card, third-party summaries) agrees the model weights are under modified AI Pubs Open RAIL-M. The HuggingFace metadata literally tags it "openrail." No credible source disputes this.

3. **Source quality?** Primary source (the actual license file in the official datalab-to/surya repo) plus the official README and official HuggingFace model card. This is the highest possible quality for a license claim.

4. **Outdated?** No. As of the current README (2025-2026 state, surya-ocr 0.17.1), the weights are still under modified AI Pubs Open RAIL-M with a revenue/funding threshold (now stated as $5M). The license remains restrictive. Note: one search snippet said "$2M" and "code is GPL," but the current authoritative sources say "$5M" and "code is Apache 2.0." These discrepancies are about peripheral details (threshold amount, code license) and do NOT touch the claim's substance ? the weights license is the point, and it is consistently OpenRAIL-M (restricted).

5. **Marketing/cherry-picked?** No. This is a factual license reading from the canonical license file.

The claim is accurate. OpenRAIL-M is unambiguously NOT in the permissive set {MIT, BSD, Apache-2.0, ISC, HPND}. It carries both a commercial revenue/funding ceiling and use-based behavioral restrictions, so it fails muneem's hard constraint #3 (permissive weights). The conclusion that Surya's weights are non-compliant follows directly.

I could not refute this claim. It is well-supported, current, and backed by primary sources.

13. user (2026-05-31T04:25:09.726Z)

Structured output provided successfully

14. assistant (2026-05-31T04:25:15.588Z)

Verification complete. The claim is well-supported and I could not refute it (refuted=false, high confidence).

Summary of findings:
- The primary source ? the MODEL_LICENSE file in datalab-to/surya ? verbatim matches the claim's quote: "AI PUBS OPEN RAIL-M LICENSE (MODIFIED), Version 0.1, March 2, 2023."
- Corroborated by the official README (raw) and the HuggingFace model card, which lists the license metadata as "openrail" and states the weights are "free for research, personal use, and startups under $5M funding/revenue" with paid commercial licensing beyond that.
- OpenRAIL-M carries both a commercial revenue/funding ceiling and use-based behavioral restrictions, placing it categorically outside the permissive set {MIT, BSD, Apache-2.0, ISC, HPND}. Surya's weights therefore fail muneem's hard constraint #3.

One peripheral note for the broader report (not affecting this claim): a stale snippet described Surya's CODE as GPL with a $2M threshold; current authoritative sources state the code is Apache 2.0 and the threshold is $5M. The claim under review concerns the WEIGHTS license, which it gets right.